RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

Subject – AWS Lab

Experiment – 6

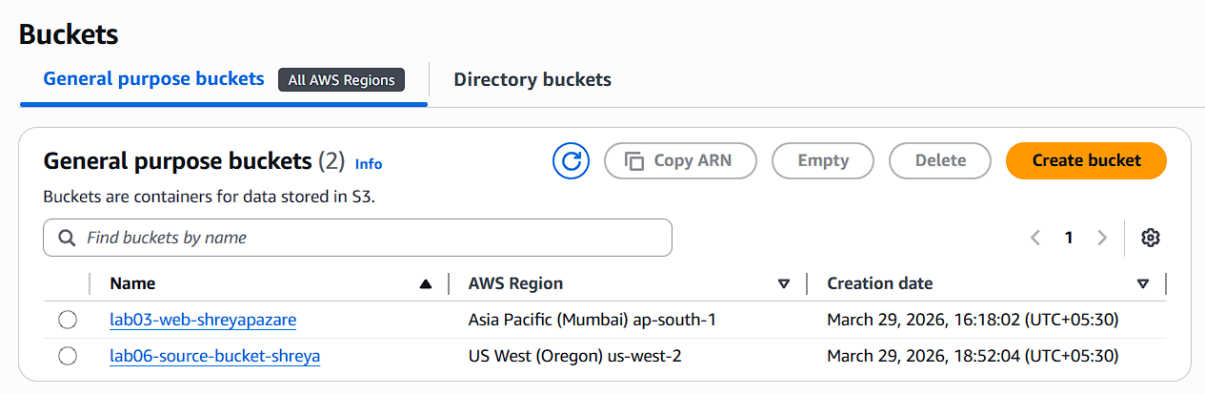
Title: Event-Driven Automation with AWS Lambda

Lab Objective:

To create a serverless function that automatically triggers whenever a new file is uploaded to an S3 bucket, demonstrating the power of event-driven architecture.

Step 1: Create an S3 "Source" Bucket

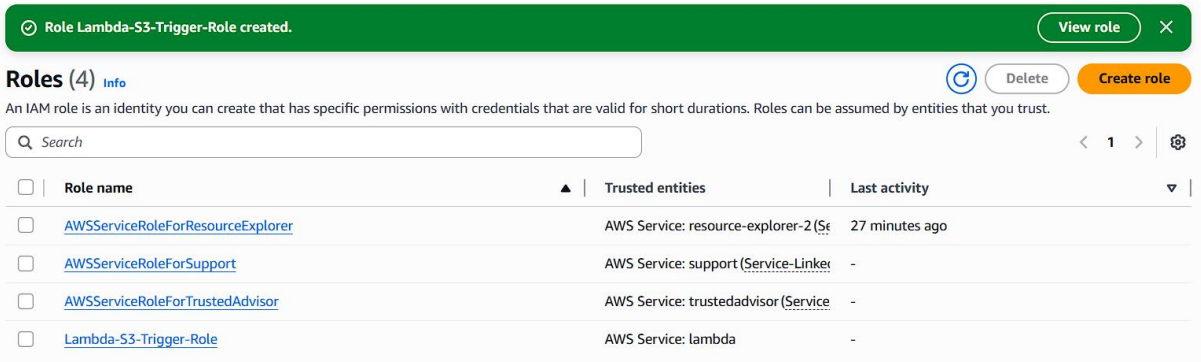
1. Navigate to the S3 Console.
2. Click Create bucket.
3. Bucket Name: lab06-source-bucket-yourname (must be unique).
4. Keep all other settings as default and click Create bucket.



Step 2: Create an IAM Role for Lambda

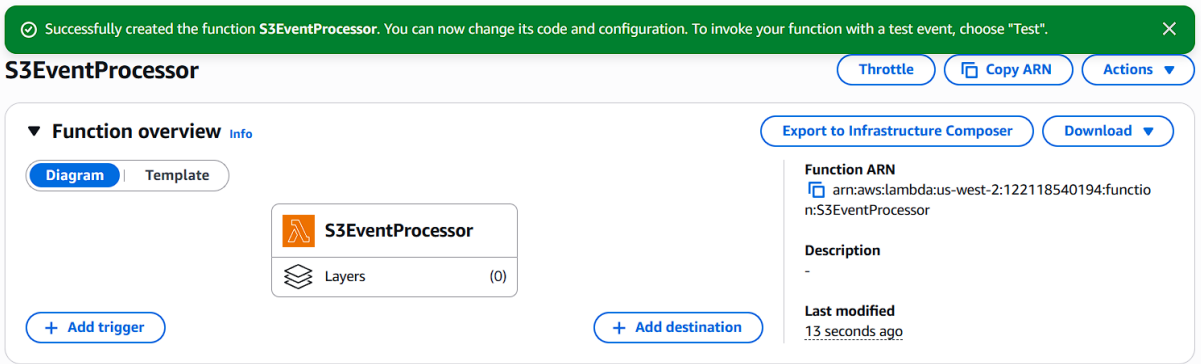
Lambda needs permission to "talk" to S3 and write logs to CloudWatch.

1. Navigate to the IAM Console > Roles > Create role.
2. Select AWS Service and choose Lambda.
3. Search for and attach these two policies:
    - AWSLambdaBasicExecutionRole (for CloudWatch logging).
    - AmazonS3ReadOnlyAccess (to read file details from the bucket).
4. Name the role: Lambda-S3-Trigger-Role and click Create role.



Step 3: Create the Lambda Function

1. Navigate to the Lambda Console > Create function.
2. Choose Author from scratch.
3. Function Name: S3EventProcessor.
4. Runtime: Choose either Python 3.12 or Node.js 20.x.
5. Permissions: Expand "Change default execution role" and select Use an existing role (Lambda-S3-Trigger-Role).
6. Click Create function.



Step 4: Add the Code & Trigger

The Code

Replace the default code in the Code Source tab with one of the following:

Python Option:

```
import json

import urllib.parse
```

```
def lambda_handler(event, context):
    # Get the bucket name and object key from the event
    bucket = event['Records'][0]['s3']['bucket']['name']
    key = urllib.parse.unquote_plus(event['Records'][0]['s3']['object']['key'],
encoding='utf-8')
    print(f"File Uploaded! Bucket: {bucket}, File Name: {key}")
    return {
        'statusCode': 200,
        'body': json.dumps('Event Processed Successfully!')
    }
```

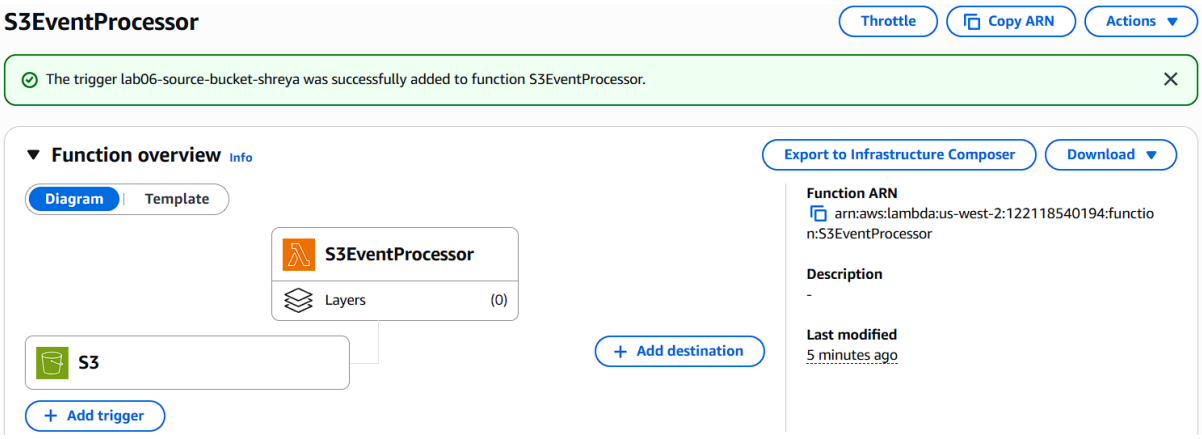
Node.js Option:

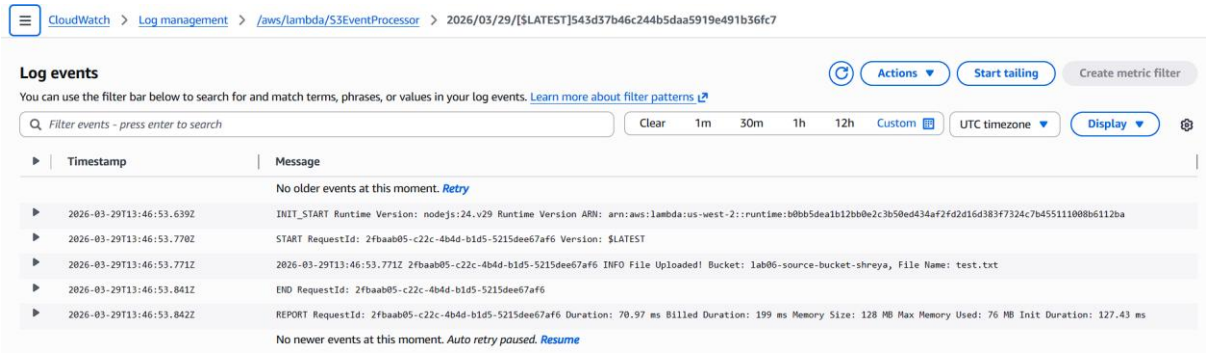
```
exports.handler = async (event) => {
    const bucket = event.Records[0].s3.bucket.name;
    const key = decodeURIComponent(event.Records[0].s3.object.key.replace(/\+/g, '
'));
    console.log(`File Uploaded! Bucket: ${bucket}, File Name: ${key}`);
    return {
        statusCode: 200,
        body: JSON.stringify('Event Processed Successfully!'),
    };
};
```

Click Deploy after pasting the code.

The Trigger

1. In the function overview, click + Add trigger.
2. Select S3 from the dropdown.
3. Select your lab06-source-bucket-yourname.
4. Event type: "All object create events".
5. Acknowledge the "Recursive invocation" warning and click Add.



Step 5: Testing & Verification

1. Go to your S3 bucket and Upload any small file (e.g., test.txt).
2. Go back to your Lambda function and click the Monitor tab.
3. Click View CloudWatch logs.
4. Open the latest log stream; you should see the message: "File Uploaded! Bucket: lab06..., File Name: test.txt".